

Briefing Document: Entity–Relationship Modeling and Relational Database Conversion

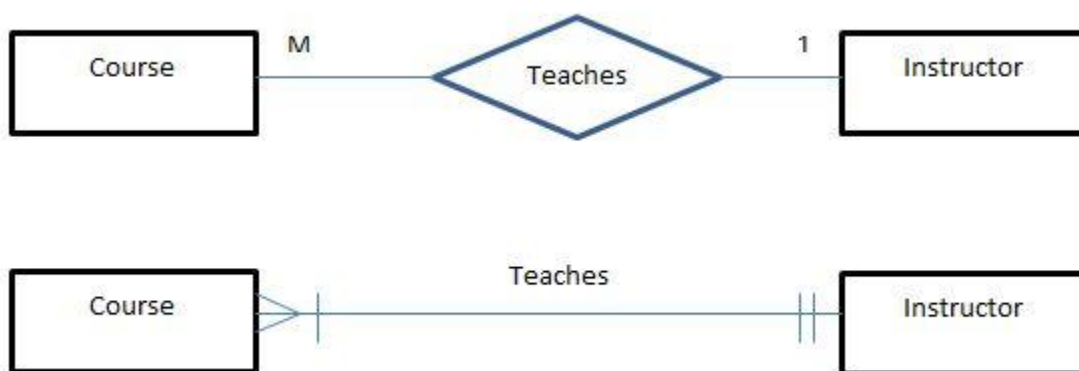
Summary

This document presents a structured explain of:

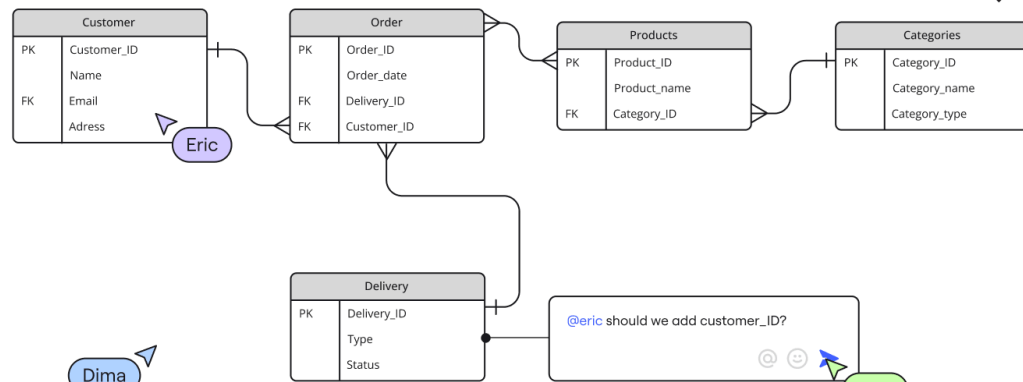
- **ER → RDBMS conversion rules**
- Effects of **cardinality and participation constraints**
- Structural treatment of **weak and recursive entities**
- Foundations of **relational query languages**
- Core operators of **Relational Algebra (RA)** and query execution logic

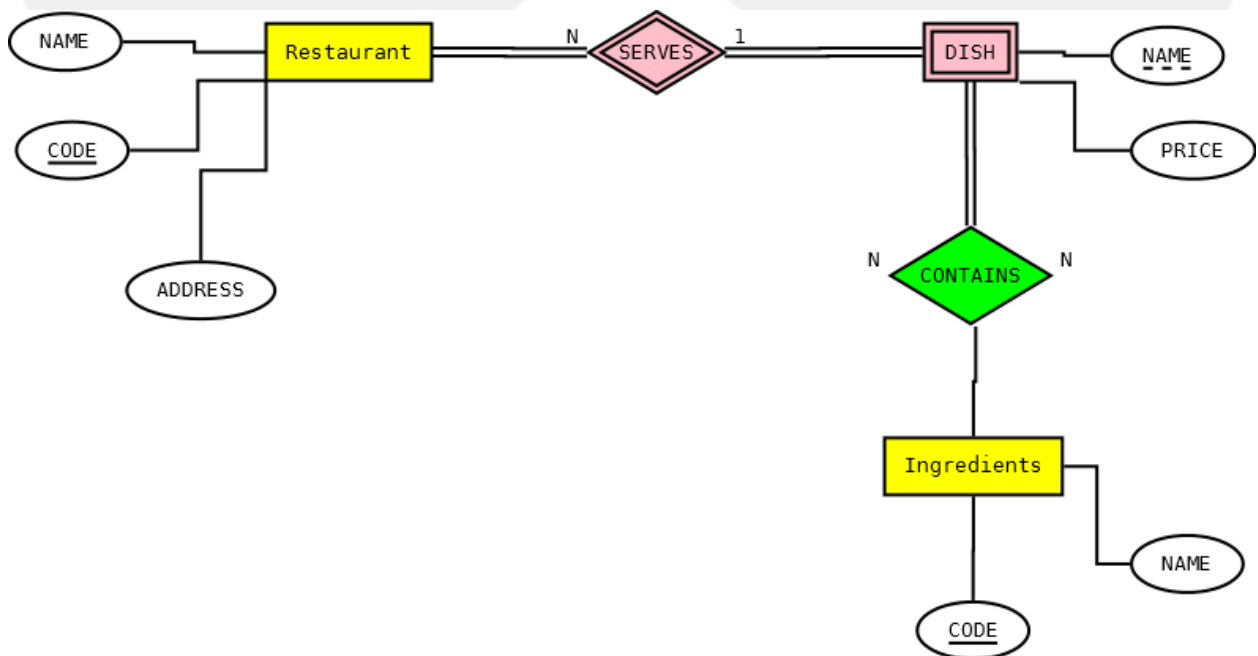
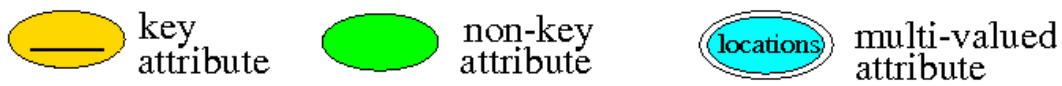
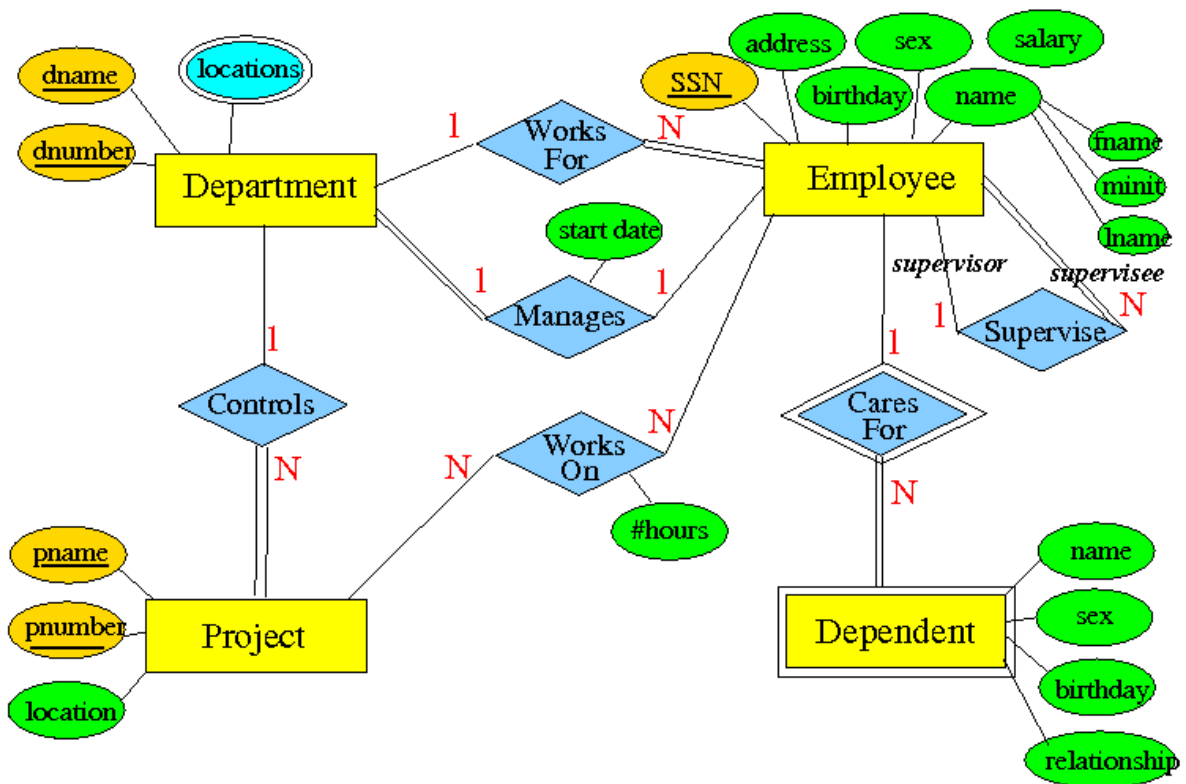
The focus is conceptual clarity + exam-oriented precision, especially for schema design and table-count questions.

1. ER Model to RDBMS Conversion



ERD e-commerce





Conversion transforms the **conceptual ER diagram** into **relational tables** using deterministic mapping rules.

Core Principle

Cardinality + Participation determine

1. Number of tables
2. Placement of foreign keys
3. Whether relationship tables are needed

1.1 Mapping Cardinality and Table Count

Cardinality	Tables Required (Partial Participation)	Foreign Key Placement
1 : 1	2	Either side
1 : M	2	On “Many” side
M : 1	2	On “Many” side
M : M	3	Separate relation table

Key Rule for 1:M and M:1

Migration always occurs toward the **Many side**.

The primary key of the “One” side becomes a **foreign key** in the “Many” side.

Example:

Student (RollNo PK)

Course (CourseID PK)

Enrollment (Student-RollNo FK, CourseID FK) ← Only for M:M

2. Participation Constraints and Table Reduction

Participation constraints significantly affect **table reduction**.

2.1 Total Participation on Both Sides

If **both sides are total**, then:

- Relationship can often be merged
- Final result may reduce to **1 table**

Applies to:

- 1:1
- 1:M
- M:1

△ Exception: M:M generally still needs separate relation unless attributes and constraints allow safe merging.

2.2 Total Participation on One Side

Case	Result
Total on “One” side	Tables reduce to 1
Total only on “Many” side	No reduction (remain 2)

Why?

If the “One” side has total participation, every row must relate — so merging is safe.

If only the “Many” side is total, merging causes redundancy or nulls.

3. Specialized Entity Types

3.1 Strong vs Weak Entities

Strong Entity

- Independent
- Has its own primary key

Weak Entity

- Depends on strong entity
- Double rectangle
- Has **partial key**
- Participation is always **total**

Identity Formation Rule

Primary Key (Weak) = PK(Strong) + Partial Key

Example:

Employee (EmpID PK)

Dependent (EmpID FK + DependentName) → Composite PK

3.2 Recursive Relationships

Recursive = entity relates to itself.

Table Count Rules

Recursive Type	Tables
1:1	1
1:M or M:1	1
M:M	2

For 1:M recursive:

- Add a self-referencing foreign key

Example:

Employee(EmpID PK, ManagerID FK referencing Employee)

4. Relational Query Languages

Relational query languages operate at the logical level.

4.1 Procedural vs Non-Procedural

Type	Example	Specifies
Procedural	Relational Algebra	What + How
Non-Procedural	SQL, TRC, DRC	What only

Relational Algebra

- Theoretical foundation
- Used for query optimization theory
- Basis for SQL execution planning

5. Relational Algebra Operators

5.1 Basic Operators

1. Selection (σ)

- Horizontal filtering
- Returns rows satisfying condition

Example:

$\sigma \text{ CGPA} > 9 \text{ (Student)}$

Returns full tuples where $\text{CGPA} > 9$.

2. Projection (π)

- Vertical filtering
- Returns selected attributes only

Example:

$\pi \text{ Name (Student)}$

Returns only Name column.

Other Basic Operators

- Union
- Set Difference
- Cross Product
- Rename (ρ)

5.2 Derived Operators

Built from basic operators:

- Join
- Intersection
- Division

Example:

$\text{Student} \bowtie \text{Enrollment}$

6. Query Logic and Execution Order

Order of operations is critical in relational algebra.

Golden Rule

Selection should generally precede Projection.

Why?

Projection removes columns.

If projection happens first:

- Required attributes for selection may be lost.

Correct Order Example

To find names of students with CGPA > 8:

π Name (σ CGPA > 8 (Student))

Step-by-step:

1. Select rows where CGPA > 8
2. Project Name from filtered result

If reversed:

σ CGPA > 8 (π Name (Student))

✗ Fails because CGPA column no longer exists.

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